Mod 3

Regularization

Regularization is a technique used to prevent overfitting.

1) L1 (Lasso) Regularization)

LASSO - Least absolute shrinkage and Selection operator.

$$\text{cost} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\sum_{i=1}^{m}|w_i|$$

m = no. of features λ = parameter
n = no. of examples
y_i = actual target value
$\hat{y}_i$ = predicted target value

2) Ridge Regression (L2)

$$\text{cost} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\sum_{i=1}^{m}w_i^2$$

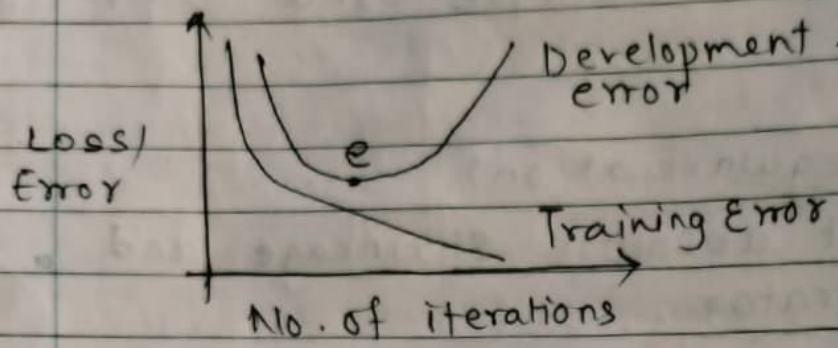
3) Dropout

Ignores some layer outputs to prevent overfitting.

1) During training, a chosen proportion of neurons are randomly deactivated.
2) Remaining active neurons are scaled up.

4) Early stopping

We stop training on model when performance on validation set is getting worse.

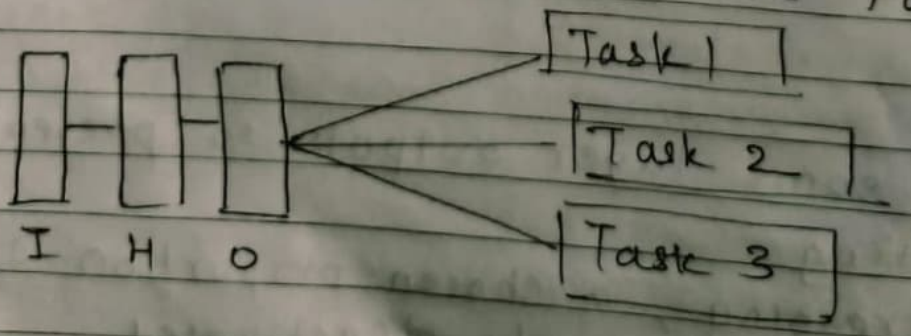


e = Early stopping

- Multi-Task Learning.

MTL is a type of deep learning technique where a model is trained to perform multiple tasks simultaneously.

The goal is to improve the generalization performance of the model by leveraging the information shared across tasks.



Applications - Object dection, NLP, Self driving cars.

- Noise Robustness / Parameter Norm Penalties

Noise applied to inputs is data augmentation. For some models, addition of noise with extremely small variance at the input is equivalent to imposing a penalty on the norm of the weights.

Noise applied to hidden units can be much more powerful than the parameters.

This is parameter norm penalty. This can be interpreted as a stochastic implementation of Bayesian inference over the weights.

- Parameter Typing

Two models are doing the same classification task but their input distributions are different.

- Parameter Sharing

The parameters of one model, trained as a classifier in a supervised paradigm are regularised to be close to the parameters of another model.

- Bagging - Bootstrap Aggregating is designed to improve the stability and accuracy of algorithms used in statistical classification and algorithms.

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Working

1) Create multiple subsets from original data with equal tuples.
2) A base model is created on each subset
3) Each model is learned in parallel
4) Final predictions are determined by combining all models.

Challenges in NN Optimization

1) Ill conditioning - poor Hessian matrix
2) Local minima - problem in finding local minima
3) Cliffs - having steep layers
4) Long Term dependencies - graphs become extremely deep.
5) Theoretical Limits of optimization